VIHANGA THALANGEDARA



Software Engineer

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PROFILE SUMMARY

Software Engineer, passionate about **solving real-world problems** through scalable and user-centered software solutions. Bringing hands-on experience in **full-stack development,** specializing in **React.js, Node.js, Python,** with expertise in **JavaScript** and **modern UI frameworks,** and practical experience in **MySQL , MongoDB** and **cloud** concepts**.** Solid understanding of **Agile** methodologies, **software design principles** and **best practices** through **industry internship** experience. A **collaborative team player** with a strong **learning mindset**, committed to **continuous improvement** and delivering **high-quality** solutions.

WORK EXPERIENCE

Associate Software Engineer Intern - Adveccio (Pvt) Ltd. *MAY 2024 - NOV 2024*

- **Web Application Rewrite:** Migrated a system to a modern LAN-based web application by redesigning and implementing **user-focused frontend features** while integrating **JWT authentication** and **role-based access control**, ensuring secure data handling aligned with industry best practices.
- Designed and developed **7+ dynamic and responsive UI** screens using **React Hooks, Context API** for state management, and **reusable component architecture** and **custom theme integration** feature**.**
- Implemented **lazy loading** to improve initial load performance (**~30% faster**) and optimized rendering using **React.memo, useMemo,** and **useCallback** for enhanced application performance.
- Developed and integrated secure **RESTful APIs** using Flask, implementing **ORM** with **SQLAlchemy** for efficient database operations and structured **data modeling** with **MySQL**.
- Improved existing dashboard visualizations by **redesigning chart presentation** and enhancing data clarity.
- **ERP System:** Collaborated in a **6-member Agile team** to develop report generation modules and **resolve 5+ critical bugs,** including handling unsaved data scenarios to improve **data integrity**.
- Contributed to **backend API** enhancements and maintained **clean Git/GitHub workflows** following structured **debugging** and **testing practices.**
- **Manufacturing Web Application:** Built a data-driven web application to analyze the financial impact of manufacturing delays with insightful dashboards.
- Integrated **RESTful API** endpoints, and implemented **data visualization** using **Chart.js** and contributed to performance tuning and **frontend optimization** to improve **responsiveness** and scalability.
- Technologies Used - JavaScript, React.js, Bootstrap, Tailwind CSS, Flask, MySQL, RESTful APIs, Git/GitHub

EDUCATION

BSc. Engineering (Hons) in Computer Engineering - University Of Sri Jayewardenepura *2021 - 2025*

- Second Class Honours - GPA - 3.2
- Minor in Data Management

PROJECTS

Seatify - Real-Time Cafe Occupancy Monitoring Web App (Final Year Project) *NOV 2024 - JULY 2025*

- Initiated and led the project concept using CCTV feeds and computer vision for real-time seat occupancy detection, reducing wasted visits for customers and helping cafe owners manage seating efficiently.
- Developed key features including responsive UI, reusable components, theme integration, secure authentication, role-based access, RESTful API integration while delivering insightful dashboards with charts.
- **Selected for the Final Rounds of NBQSA ICT Awards 2025 under the Tertiary Business Category.**
- Tech Stack: React.js, Flask, Tailwind CSS, MySQL, JWT authentication, RESTful APIs

VidHub - Team-Based Video Resource Management Platform (Individual Project) *DEC 2025 - JAN 2026*

- Built a full-stack web application to solve the problem of fragmented video knowledge sharing within teams by providing a centralized platform for curating, organizing, and managing YouTube video contents.
- Handled state management with **React Hooks** for smoother user interaction, while implementing **team-based access control** and MongoDB with **ODM** to ensure structured data modeling and isolation.
- Tech Stack: React.js, Node.js, Express.js, Material UI, MongoDB, JWT Authentication, RESTful APIs

Interacta - AI-Powered Dynamic Quiz Generator (Individual Project) *JULY 2025 - AUG 2025*

- Developed an AI-powered web application that automatically generates quizzes from product descriptions or arbitrary content, reducing manual quiz creation and increasing interactive customer engagement.
- Developed **4 responsive UIs** with **reusable components** and structured layouts to ensure consistency across multiple views while integrating **asynchronous RESTful APIs** for dynamic quiz creation and management.
- Tech Stack: React.js, Node.js, Express.js, Bootstrap, MongoDB, LLM API Integration, RESTful APIs

TECHNICAL SKILLS

Skills: JavaScript, Python, React.js, HTML, CSS, Node.js, Express.js, Flask, Bootstrap, Tailwind CSS, Material UI, SQL, MongoDB, MySQL, bcrypt, RESTful APIs, Zustand, Context API

Cloud: Microsoft Azure, Vercel, Render

Tools and Platforms: Git/GitHub, Postman, Figma, Taiga, Microsoft SQL Server

EXTRA-CURRICULAR ACTIVITIES

Graphic Designer *2022 - 2025*

- CENSOC Computer Engineering Society & Technical Sub-Committee - FOE, USJ

Deputy Head Prefect *2017/2018*

- Gurukula College, Kelaniya

Secretary - Science Society *2017/2018*

- Gurukula College, Kelaniya

House Captain - Inter House Sports Meet *2017/2018*

- Gurukula College, Kelaniya

CERTIFICATIONS

- React Basics - Meta (Coursera) *OCT 2025*
- Programming with JavaScript - Meta (Coursera) *DEC 2025*
- Introduction to Front-End Development - Meta (Coursera) *JULY 2024*
- MySQL Explorer Badge - Oracle *MAY 2025*
- Getting Started with Git and GitHub - IBM (Coursera) *SEPT 2025*
- Python for Beginners - University of Moratuwa *JULY 2022*
- Google AI Essentials Specialization - Google (Coursera) *SEPT 2025*

SOFT SKILLS

Leadership, Team Collaboration, Quick Learning, Adaptability, Effective Communication, Creative Problem Solving

INTERESTS

UI/UX, Painting, Reading, Graphic Designing, Coding, Yoga, Meditation

REFERENCES

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